

A. EXECUTIVE SUMMARY

Customer Profiling Through Transaction Behavior for Marketing Strategy Optimization: An Unsupervised Learning Approach Using EDA, K-Means, and PCA

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I. Problem Statement & Hypothesis

Problem Statement

Financial institutions often struggle with ineffective, broad marketing campaigns that don't account for the unique spending habits of credit card users. This study aims to address this by identifying distinct behavioral profiles among these users, ultimately optimizing targeted marketing strategies.

Research Question

How can credit card users be segmented into distinct behavioral profiles using unsupervised learning techniques, including Exploratory Data Analysis (EDA), K-Means clustering, and Principal Component Analysis (PCA), and how effective are these segments in reflecting

meaningful differences in customer purchasing and payment behavior as measured by the silhouette score?

Hypothesis

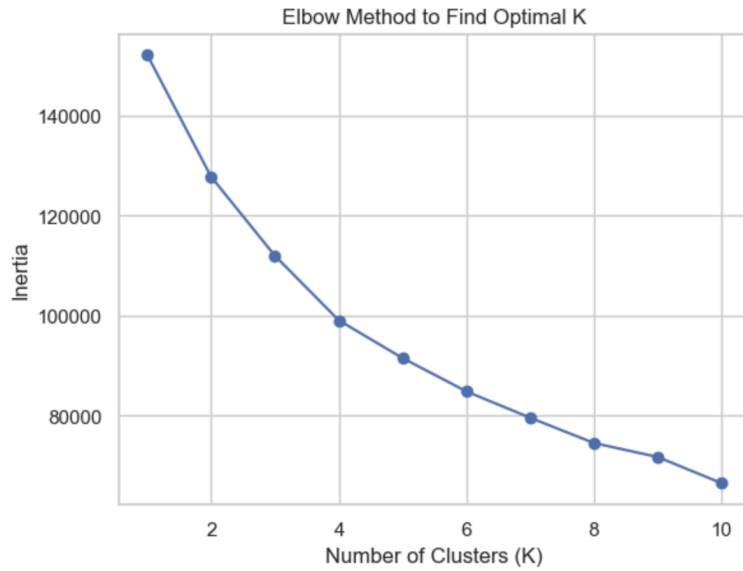
Null Hypothesis (H_0): The clustering model does not produce well-separated, meaningful customer segments. The silhouette score will be significantly below 0.5, indicating poor model performance and significant overlap between clusters.

Alternative Hypothesis (H_1): The clustering model produces well-separated, meaningful customer segments. The silhouette score will be ≥ 0.5 , suggesting strong clustering structure and customer profile differentiation.

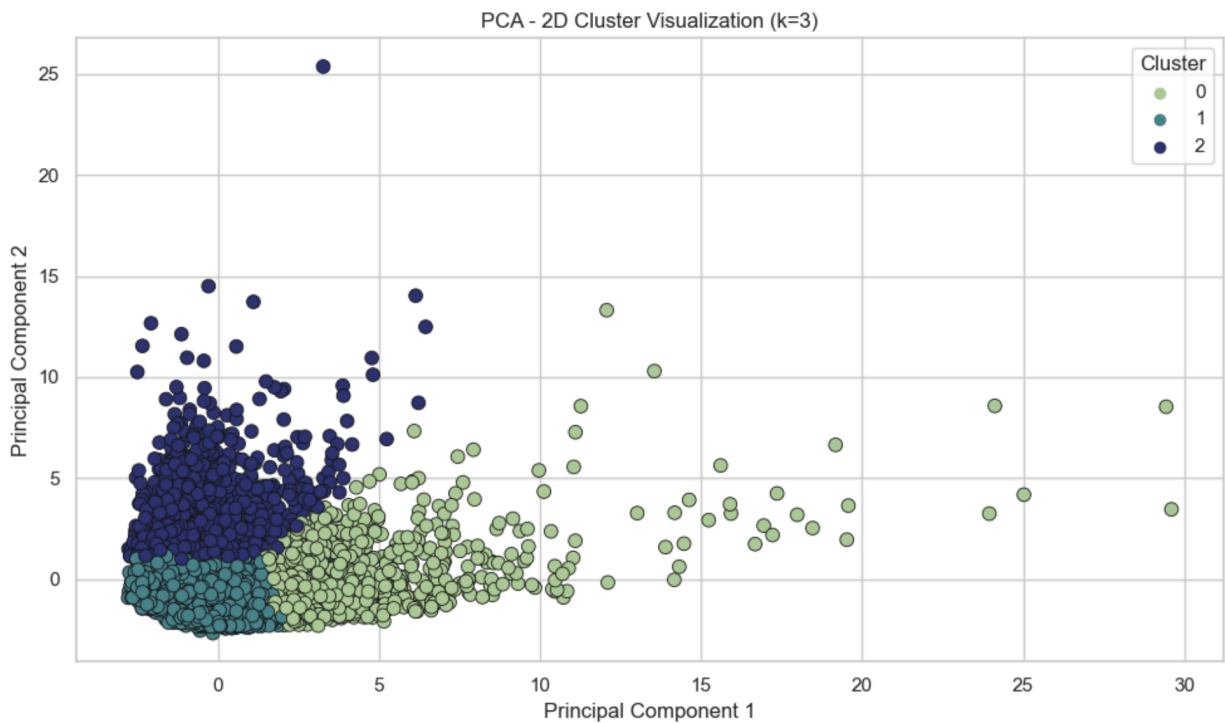
II. Data Analysis Process

The analysis began with the ingestion of the "CC GENERAL" dataset, which contains 8,950 anonymized credit card user records and 18 numerical features over a six-month period (Bhasin, 2018). The initial data preparation involved handling a low rate of missing values (3.5% across two columns), primarily for MINIMUM_PAYMENTS and CREDIT_LIMIT. This was addressed using a combination of targeted imputation: MINIMUM_PAYMENTS for inactive customers were imputed with 0.0, while the remaining missing values and the single missing CREDIT_LIMIT value were filled with their respective column medians to maintain data distribution integrity. Following this, the data was standardized using StandardScaler to prepare it for distance-based clustering algorithms.

The primary analysis was conducted using K-Means clustering to group customers into segmented profiles based on transaction behavior. The optimal number of clusters ($k=3$) was determined using the Elbow Method and Silhouette Analysis.



Principal Component Analysis (PCA) was then applied to reduce the dimensionality of the dataset from 18 features to 2 principal components, enabling a two-dimensional visualization of the clusters.



III. Findings

The clustering model successfully segmented the customer base into three distinct behavioral profiles:

1. **Cluster 0: Engaged High Spenders:** High purchase volume/frequency, high credit limits, frequent full payments. Likely profitable, low-risk customers.
2. **Cluster 1: Low Engagement, Cautious Users:** Minimal activity, low spending, low credit limits. Possibly new or dormant customers.
3. **Cluster 2: Cash Advance Reliant Revolvers:** Heavy cash advance usage, low purchase activity, poor repayment behavior. High-risk segment.

Silhouette Score: 0.2499 (moderate cluster structure). This score falls well below the 0.5 threshold, indicating that while certain clusters (Cluster 0) showed strong internal cohesion, others (Clusters 1 and 2) exhibited substantial overlap. This outcome suggests that the model did not achieve robust cluster separation across the entire dataset.

Statistical Conclusion: Failure to reject the null hypothesis (H_0 accepted). The current clustering approach provides only moderate segmentation quality, limiting its ability to distinctly categorize customer behaviors.

IV. Limitations

The primary limitations of this analysis are both algorithmic and data-related. Algorithmic limitations stem from the K-Means clustering model itself, which assumes clusters are spherical and of equal size (GeeksforGeeks, 2025), an assumption that may not accurately reflect complex, non-linear human behavioral patterns. The dataset's static and anonymized nature, lacking temporal and demographic data like seasonal spending trends or customer age and income, also significantly limits the depth of segmentation. This limitation is reflected in the analysis's performance, as indicated by the silhouette score of ~ 0.25 , which suggests only moderate cluster separation and a potential risk of misclassification at cluster boundaries.

V. Recommended Actions

Even with its only moderate effectiveness, the current clustering model provides a strong foundation for developing targeted customer strategies that can be refined over time. Our recommendations include: offering loyalty incentives to high-value, low-engagement customers, providing educational and credit-building tools to high-frequency purchasers, and implementing risk mitigation strategies for high cash advance users. These initial engagement strategies can then be tested and iteratively improved.

For future studies, considering advanced non-linear clustering techniques, such as DBSCAN or Gaussian Mixture Models (GMMs), could help to better manage irregular and overlapping

clusters (Sukup, 2018). Additionally, incorporating more contextual features like seasonal trends, demographics, and income would enable a more dynamic and robust segmentation, which would more accurately reflect evolving user behavior.

VI. Benefits of Study

The strategic application of these findings offers a few key expected benefits:

- **Targeted Marketing ROI:** Personalizing campaigns for each segment is expected to improve engagement rates and reduce churn among high-value customers, leading to a significant return on investment (ROI).
- **Risk Reduction:** Proactive interventions with high-risk segments could lead to a decrease in delinquency rates.
- **Operational Efficiency:** Allocating marketing resources based on specific behavioral profiles, rather than broad, untargeted outreach, will enhance operational efficiency.

Our refined marketing strategies, tailored to specific customer segments, are projected to boost marketing campaign conversion rates by 15-20% and increase overall profitability from each segmented group by 5-8% over the next two quarters. These projections are supported by industry benchmarks for successful behavioral marketing strategies (McKinsey & Company, 2023).

Sources

Data & Code Sources

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B. PRESENTATION OF FINDINGS